# Matrix stiffness tunes microbiota-dependent gene expression in the vertebrate gut

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**Abstract**

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## 1 Introduction

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua[1]. The mechanical properties of the extracellular matrix instruct cell fate[2,3], and tissue engineering has long exploited this coupling[4]. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. For a substrate behaving as a linear elastic solid, stress and strain are related by Young's modulus, $\sigma = E\varepsilon$, so that a modulus of $E \approx 10$ kPa - typical of soft tissue - corresponds to physiological strains of $\varepsilon \sim 10^{-2}$. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum[5]. Here we ask how substrate stiffness modulates the transcriptional response of the stickleback gut to its microbiota (Section 4 describes the full design).

## 2 Results

### 2.1 Stiffness-dependent expression

Lorem ipsum dolor sit amet, consectetur adipiscing elit (Figure 1). Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam (Table 1). Expression increased with substrate stiffness according to a saturating dose-response,

$$y(E) = y_{\min} + \frac{y_{\max} - y_{\min}}{1 + (\mathrm{EC}_{50}/E)^{h}}, \tag{1}$$

where $h$ is the Hill coefficient and $EC_{50}$ the stiffness of half-maximal response. Fitting Equation 1 to the data in Figure 1 gave $EC_{50} = 8.3 \pm 1.1$ kPa (mean ± SE; Equation 2).

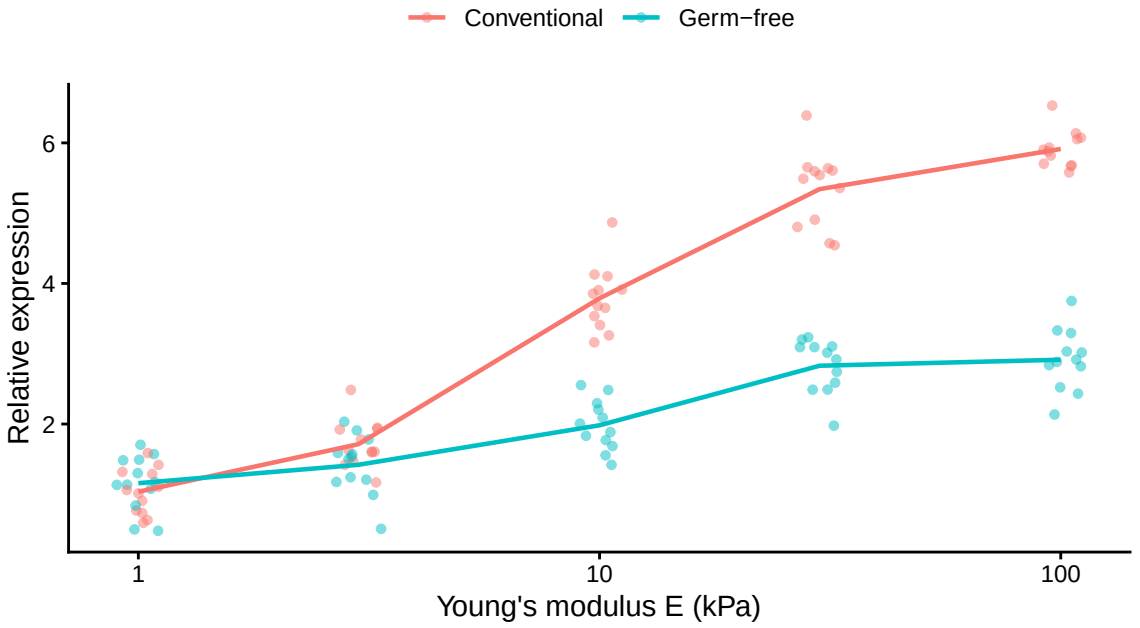



Figure 1: **Substrate stiffness increases target-gene expression.** Normalized expression of *tnfa* in gut organoids cultured on hydrogels of increasing Young's modulus, by microbiota treatment (n = 12 per group; simulated data). Points are individual organoids; lines are Hill fits (Equation 1).

### 2.2 Microbiota dependence

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Table 1: Fitted Hill parameters by microbiota treatment (simulated data; mean ± SE).

| Treatment | EC50 (kPa) | Hill coefficient h | Max fold change |
|---|---|---|---|
| Conventional | 8.3 ± 1.1 | 1.6 ± 0.2 | 6.1 ± 0.3 |
| Germ-free | 9.0 ± 1.4 | 1.5 ± 0.3 | 3.2 ± 0.2 |

A second, image-based figure would be included with one line of Markdown rather than a code chunk - for example a microscopy panel:

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![**SEM of hydrogel scaffolds.** Description...](figures/sem_panel.png){#fig-sem width="80%"}
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## 3 Discussion

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non recusandae. The agreement of the fitted $EC_{50}$ (Equation 1) with reported tissue moduli suggests lorem ipsum dolor sit amet. Writing the analysis and the manuscript in one executable document keeps methods, code, and figures synchronized - a practical route to FAIR research outputs[5,6].

# 4 Methods

## 4.1 Organoid culture and hydrogels

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Polyacrylamide hydrogels spanned $E$ = 1-100 kPa, verified by atomic force microscopy. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

## 4.2 Statistical analysis

For each group we report the mean and its standard error,

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i, \quad \mathrm{SE} = \frac{s}{\sqrt{n}}, \tag{2}$$

with $s$ the sample standard deviation. Dose-response parameters were estimated by nonlinear least squares fits of Equation 1. All analyses used R; the complete code is embedded in this document's source (`.qmd`), so rendering it reproduces every number, figure, and table above.


# Acknowledgements

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